

Finite-Mesh Certification of Continuous Resolvent Contours

A Banach-Lemma Envelope for Nonnormal Packet Blocks

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Abstract

Physical interface R requires a supremum of a nonnormal resolvent on a continuous contour. A finite eigenvalue plot or a list of sampled singular values does not certify that supremum. We prove a finite covering theorem. If every z on the contour lies within h_k of some sample z_k , and

$$\|(z_k - T)^{-1}\| \leq m_k, \quad h_k m_k < 1,$$

then on the associated cell

$$\|(z - T)^{-1}\| \leq \frac{m_k}{1 - h_k m_k}.$$

For n equally spaced points on a circle of radius s , the exact covering radius is $2s \sin(\pi/(2n))$. Applying the theorem to both diagonal blocks turns the Schur feedback gate into a finite-input certificate. A dense-grid audit of 256 nonnormal matrices has no envelope failure. Its floating-point sample inverses are diagnostics; validated physical inverse bounds remain an open input.

1 A local resolvent covering lemma

Let T be a bounded operator and suppose $z_k \in \rho(T)$. For any z ,

$$z - T = (I + (z - z_k)(z_k - T)^{-1})(z_k - T).$$

Theorem 1 (Finite-mesh resolvent envelope). *Let Γ be covered by sets U_k such that*

$$\sup_{z \in U_k \cap \Gamma} |z - z_k| \leq h_k, \quad \|(z_k - T)^{-1}\| \leq m_k.$$

If $h_k m_k < 1$ for every cell, then $\Gamma \subset \rho(T)$ and

$$\sup_{z \in \Gamma} \|(z - T)^{-1}\| \leq \max_k \frac{m_k}{1 - h_k m_k}.$$

Proof. For $z \in U_k \cap \Gamma$, the perturbation $(z - z_k)(z_k - T)^{-1}$ has norm at most $h_k m_k < 1$. The Neumann series inverts the first factor and bounds its inverse by $(1 - h_k m_k)^{-1}$. Multiplication by the sample resolvent proves the local bound. Taking the maximum over the finite cover proves the theorem [Kato, 1995]. $\square$

The condition is local. Adaptive refinement need only split cells where $h_k m_k$ is large; one ill-conditioned sample need not force uniform oversampling elsewhere.

2 Exact covering radius for a circle

Proposition 2 (Equal-angle circle mesh). *For $n \geq 3$ equally spaced nodes on $|z - \mu| = s$, every contour point is within*

$$h_n = 2s \sin \frac{\pi}{2n}$$

of its nearest node. This radius is sharp.

Proof. The farthest point in each arc is its midpoint. Its angular distance from either endpoint is π/n , and the corresponding chord length is $2s \sin(\pi/(2n))$. $\square$

Thus a uniform sample bound m certifies the circle whenever $2sm \sin(\pi/(2n)) < 1$. Solving this inequality supplies a deterministic minimum mesh density once m is known.

Corollary 3 (Simple mesh-complexity budget). *The sufficient integer condition*

$$n > \pi sm$$

guarantees $h_n m < 1$.

Proof. Use $\sin x \leq x$ to obtain $h_n m \leq 2sm \pi/(2n) = \pi sm/n < 1$. $\square$

The estimate is conservative but useful before sampling begins. It shows that mesh complexity grows linearly with the contour radius and the expected resolvent amplification. Near a pseudospectral spike, adaptive local cells can be much cheaper than enforcing the worst m around the entire curve.

3 Finite-input Schur certificate

Apply Theorem 1 separately to the packet and complementary blocks. Let sample inverse bounds be $m_{P,k}, m_{Q,k}$ and covering radii h_k . Define

$$a = \max_k \frac{m_{P,k}}{1 - h_k m_{P,k}}, \quad d = \max_k \frac{m_{Q,k}}{1 - h_k m_{Q,k}}.$$

Corollary 4 (Sampled packet-to-Riesz rank gate). *If all mesh products are below one and*

$$adb < 1,$$

then the full off-diagonal homotopy preserves the packet rank inside Γ .

Proof. The mesh theorem supplies continuous diagonal resolvent bounds. The two-sided Schur theorem then applies [Program, 2026]. $\square$

The directional graph formulas of RH-166 can use the same a, d . Therefore the entire finite rank-and-graph decision requires only certified sample inverses, covering radii, and two residual norms.

4 Resolvent disks and adaptive stopping

The local theorem gives an intrinsic certified neighborhood around each sample.

Proposition 5 (Certified resolvent-disk cover). *Fix $0 < \theta < 1$. A sample bound m_k certifies every point in*

$$\mathbb{D}_k = \{z : |z - z_k| \leq \theta/m_k\}$$

with the uniform local estimate

$$\|(z - T)^{-1}\| \leq \frac{m_k}{1 - \theta}.$$

If $\Gamma \subset \bigcup_k \mathbb{D}_k$, these disks form a complete continuous contour certificate.

Proof. Insert $h_k = \theta/m_k$ into Theorem 1; then $h_k m_k = \theta$. $\square$

This yields a rigorous adaptive stopping rule: add samples only on uncovered arcs, recompute their validated inverse bounds, and stop exactly when the union of certified disks covers the curve. Regions of large resolvent norm automatically receive small disks and finer sampling. Failure to terminate at a fixed precision is a conditioning diagnostic, not evidence that the contour intersects the spectrum.

5 What a sample inverse certificate requires

A numerical inverse R_k is not itself an upper bound. A standard a-posteriori route is to compute the defect

$$E_k = I - R_k(z_k - T).$$

If $\|E_k\| < 1$, then

$$\|(z_k - T)^{-1}\| \leq \frac{\|R_k\|}{1 - \|E_k\|}.$$

For a physical proof, both norms must be outwardly enclosed and the operator T must be related to its assembled matrix by a certified ball. RH-167 does not silently promote floating singular values to such bounds.

6 Dense-grid audit and route consequence

The companion audit draws 256 upper-triangular nonnormal matrices, samples a circle at 20–64 nodes, and evaluates the envelope. For every accepted mesh it compares the result with 2048 dense contour evaluations. No dense value exceeds the theorem bound. This exercises strongly nonconstant resolvents but remains a floating-point regression test.

RH-167 removes the uncountable contour quantifier from interface R. The next wall is validation: transport nominal sample inverses and couplings through an operator ball without losing the mesh denominator. No physical sample bound, uniform all-level margin, Gate A, or downstream Hilbert–Polya claim is made.

References

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.

Prime Dynamics Theory Program. A two-sided schur certificate for packet-to-riesz lifting, 2026. RH-163 manuscript.